

AWS VPC Connectivity Lab

Day 3 | October 10, 2026 | Madhu

Region: N. Virginia (us-east-1)

Objective

Build a custom VPC with public and private subnets, configure public internet routing, launch a Linux EC2 instance, and validate SSH and outbound HTTPS connectivity.

Scope and completion status

Public-subnet configuration and connectivity tests completed. Private-subnet connectivity remains pending. NAT gateway routing was discussed conceptually, but no NAT gateway was deployed.

Milestone	Result
Public network setup	Completed
SSH from Windows	Passed
Outbound HTTPS	Passed: HTTP/2 200
EC2 cleanup	Instance terminated
EBS cleanup	No volumes remained in the Region
Basic VPC networking	Retained for next week

Evidence note

This report uses the six screenshot collages supplied for documentation, the recorded walkthrough settings, and earlier cleanup screenshots. Launch settings not visible in the collages are identified as walkthrough records. Historical IPs and resource IDs are not instructions to reconnect to the terminated instance.

1. Create the custom VPC

VPC

The screenshot displays the AWS Management Console interface for VPC resources. At the top, there are buttons for 'Create VPC' and 'Launch EC2 Instances'. A note states: 'Note: Your Instances will launch in the United States region.' Below this, the 'Resources by Region' section shows 'You are using the following Amazon VPC resources' with a 'Refresh Resources' button. Two tabs are visible: 'VPCs' and 'Endpoint Services', both for the 'N. Virginia' region. The 'VPCs' tab shows a list with a 'See all regions' link. A green notification bar at the bottom of the console states: 'You successfully created vpc-04816a66930992bdf / day3-vpc-lab'. Below the notification, the details for 'vpc-04816a66930992bdf / day3-vpc-lab' are shown. The details are organized into a grid with the following information:

Details	
VPC ID vpc-04816a66930992bdf	State Available
DNS resolution Enabled	Tenancy default
Main network ACL acl-0e57553b29a019581	Default VPC No
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled
Encryption control ID -	Encryption control mode -
Block Public Access Off	DHCP option set dopt-0b738714dd8cb4ef6
IPv4 CIDR 10.0.0.0/16	Route 53 Resolver DNS Firewall rule groups -
DNS hostnames Disabled	Main route table rtb-0930d9f85373e9955
	IPv6 pool -
	Owner ID 083959627878

At the bottom of the details section, there are tabs for 'Resource map', 'CIDRs', 'Flow logs', 'Tags', 'Related resources', and 'Integrations'.

Configuration

Created day3-vpc-lab with IPv4 CIDR 10.0.0.0/16. State: Available. Default VPC: No. Tenancy: Default. DNS resolution: Enabled. DNS hostnames: Disabled.

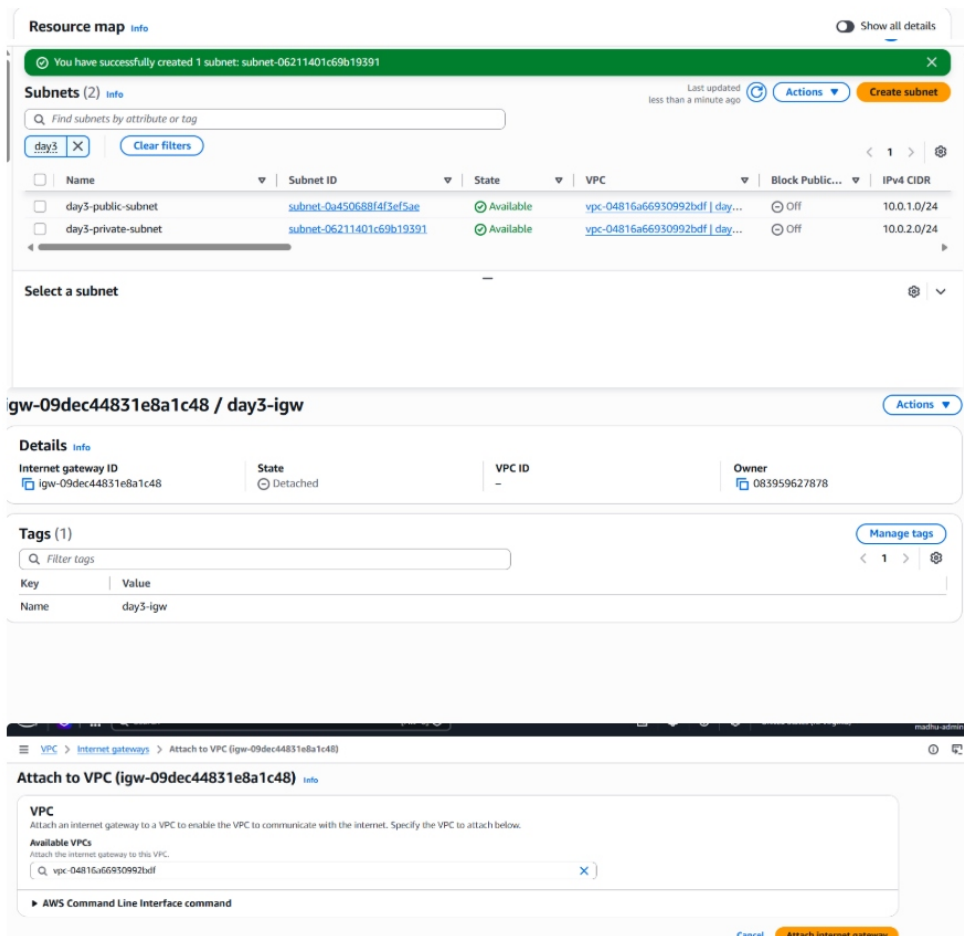
Purpose

The VPC provides the network boundary and private address space for the lab. The two subnet ranges were allocated within this CIDR.

Evidence and observation

The screenshot shows successful VPC creation. DNS resolution and DNS hostnames are separate settings; the later instance summary had no public DNS name, so SSH used the public IPv4 address.

2. Create subnets and the Internet Gateway



Subnet configuration

Created day3-public-subnet (10.0.1.0/24) and day3-private-subnet (10.0.2.0/24). Both were Available and belonged to the lab VPC.

Gateway configuration

Created day3-igw. The collage shows it initially Detached and then the attachment form with the lab VPC selected. Attachment was completed during the walkthrough; the later active route and successful internet test support the completed setup.

Key lesson

A subnet name does not make it public. A route to an Internet Gateway provides public-subnet routing. Direct IPv4 internet access for an instance also requires a public IPv4 address and appropriate security rules.

3. Configure the public route table

Create route table [info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

day3-public-rt

VPC
The VPC to use for this route table.

vpc-04816a66930992bdf (day3-vpc-lab)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q day3-public-rt X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

Updated routes for rtb-0689aecb2ed9c3f04 / day3-public-rt successfully

Details

rtb-0689aecb2ed9c3f04 / day3-public-rt Actions

Details [info](#)

Route table ID
rtb-0689aecb2ed9c3f04

VPC
vpc-04816a66930992bdf | day3-vpc-lab

Main
No

Owner ID
083959627878

Explicit subnet associations
-

Edge associations
-

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (2) Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-09dec44831e8a1c48	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Configuration

Created day3-public-rt in day3-vpc-lab. Routes: 10.0.0.0/16 -> local; 0.0.0.0/0 -> day3-igw. Both routes were Active.

Purpose

The local route handles VPC traffic. The default IPv4 route directs traffic not matched by a more specific route toward the Internet Gateway.

Evidence

The route-update success banner confirms that the internet route was saved. At this stage, the route table still needed association with the public subnet.

4. Associate the subnet and configure SSH access

The screenshot displays the AWS Management Console interface. At the top, a green notification bar states: "You have successfully updated subnet associations for rtb-0689aebc2ed9c3f04 / day3-public-rt." Below this, the "Route tables (1/3) info" section shows a table of route tables. The table has columns for Name, Route table ID, Explicit subnet associations, Edge associations, Main, and VPC. The row for "day3-public-rt" (ID: rtb-0689aebc2ed9c3f04) is selected, showing its association with "subnet-0a450688f4f3ef5ae" (10.0.1.0/24). Below the table, the "Subnet associations" tab for "rtb-0689aebc2ed9c3f04 / day3-public-rt" shows "Explicit subnet associations (1)" with the same subnet and CIDR. The bottom section, "Create security group", shows the "Basic details" for "day3-public-ec2-sg" in the "vpc-04816a66930992bdf" VPC. The "Inbound rules" section shows a rule for "SSH" (TCP) on port "22" from "My IP" (43.225.166.76/32).

Name	Route table ID	Explicit subnet associations	Edge associations	Main	VPC
-	rtb-080dc0e1f416c9f38	-	-	Yes	vpc-025b2f205b9839562
day3-public-rt	rtb-0689aebc2ed9c3f04	subnet-0a450688f4f3ef5ae...	-	No	vpc-04816a66930992bdf day3-public-ec2-sg
-	rtb-0930d9f85373e9955	-	-	Yes	vpc-04816a66930992bdf day3-public-ec2-sg

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
day3-public-subnet	subnet-0a450688f4f3ef5ae	10.0.1.0/24	-

Create security group

Basic details

Security group name: day3-public-ec2-sg

Description: Allow SSH from my IP for the Day 3 lab

VPC: vpc-04816a66930992bdf (day3-vpc-lab)

Inbound rules

Type: SSH | Protocol: TCP | Port range: 22 | Source type: My IP | Source: 43.225.166.76/32 | Description - optional:

Subnet association

Explicitly associated day3-public-subnet with day3-public-rt. The association tab shows the public subnet and its CIDR 10.0.1.0/24.

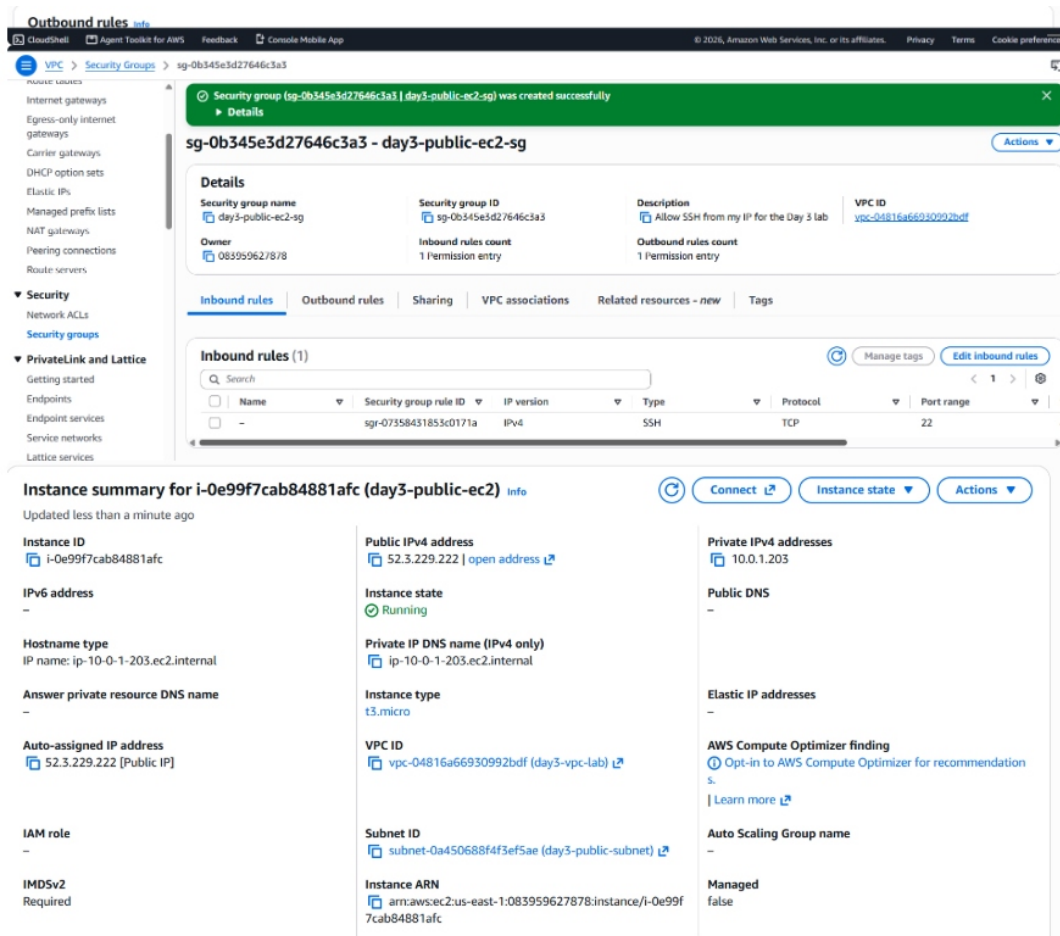
Security group

Created day3-public-ec2-sg in the lab VPC. Inbound rule: SSH / TCP / port 22 / My IP (a single IPv4 address with /32). The default outbound rule was retained during the walkthrough.

Key lessons

A route table must be associated with the intended subnet. Subnets without an explicit association use the VPC main route table. Route tables choose traffic paths; security groups control permitted traffic. SSH was not opened to Anywhere.

5. Launch the public EC2 instance



Confirmed by screenshot

day3-public-ec2 reached Running state. Instance type: t3.micro. Public IPv4: 52.3.229.222. Private IPv4: 10.0.1.203. VPC: day3-vpc-lab. Subnet: day3-public-subnet.

Recorded launch settings

Amazon Linux 2023, 64-bit x86; key pair day3-lab-key; public IP enabled; existing day3-public-ec2-sg; one 8 GiB gp3 root volume; Delete on termination: Yes; CPU credits: Standard; no additional file systems.

Key handling

Created an RSA key pair in PEM format and saved day3-lab-key.pem under C:\Users\madhu\.ssh. Keep the private key outside the Git repository. Do not include its contents in reports or screenshots.

Cost control

Standard CPU-credit mode avoids surplus CPU-credit charges, not regular instance charges. Compute, EBS storage, and public IPv4 usage were chargeable during the test. No NAT gateway was created.

6. Test SSH and outbound HTTPS

```
PS C:\Users\madhu\.ssh> ssh -i .\day3-lab-key.pem ec2-user@52.3.229.222
The authenticity of host '52.3.229.222 (52.3.229.222)' can't be established.
ED25519 key fingerprint is SHA256:SvBIB+QD7+AogggjSzn3q61Qso1vpw7WoHkmSmKoXnw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '52.3.229.222' (ED25519) to the list of known hosts.

      _#-
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Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-1-203 ~]$
[ec2-user@ip-10-0-1-203 ~]$
[ec2-user@ip-10-0-1-203 ~]$
```

```
[ec2-user@ip-10-0-1-203 ~]$ curl -I --connect-timeout 10 https://aws.amazon.com
HTTP/2 200
content-type: text/html; charset=utf-8
date: Sat, 10 Oct 2026 08:54:43 GMT
x-content-type-options: nosniff
server: Server
x-frame-options: SAMEORIGIN
x-ssr-protection: 1; mode=block
strict-transport-security: max-age=47304000; includeSubDomains
x-amz-id-1: A42013uEA64cFC8E64
last-modified: Thu, 08 Oct 2026 18:14:17 GMT
vary: accept-encoding
set-cookie: aws-privacy=J2IjovLCJldSI6MCwic3Q1OjB9; Version=1; Comment="Anonymous cookie for privacy regulations"; Domain=.aws.amazon.com; Max-Age=31536000; Expires=Sun, 10 Oct 2027 08:54:43 GMT; Path=/; Secure
set-cookie: aws_langmen; Domain=.amazon.com; Path=/
x-cache: Miss from cloudfront
via: 1.1 db129cd1d3ba1b67c3281d54986cdcc.cloudfront.net (CloudFront)
x-amz-cf-pop: IAD61-912
x-amz-cf-id: VCwXFXyM8onxKxMaLFVok6BQ8kvFYwYBfO_LDMS_TXK1WgwCZKvrA==
```

Windows preparation

Opened PowerShell, changed to the .ssh folder, and verified the key with: dir day3-lab-key.pem. The Linux command ls -ltrh failed because PowerShell ls is an alias for Get-ChildItem, which does not accept those flags.

SSH command

```
ssh -i .\day3-lab-key.pem ec2-user@52.3.229.222
```

SSH result

The Amazon Linux 2023 banner and [ec2-user@ip-10-0-1-203 ~]\$ prompt appeared. SSH authentication and access succeeded.

HTTPS command

```
curl -I --connect-timeout 10 https://aws.amazon.com
```

HTTPS result

Received HTTP/2 200. The instance successfully reached that HTTPS destination. The -I option requests response headers only. This is not proof of private-subnet connectivity or all internet protocols.

Security lesson

The first-connection host fingerprint was accepted without a documented independent check. In future, compare it with the fingerprint in the instance system log before accepting.

7. Cleanup and retained resources

Exited SSH and terminated day3-public-ec2. An earlier screenshot confirmed the state Terminated. The Volumes page then reported no volumes in the Region, confirming the lab disk was removed.

Delete on termination was set to Yes for the root EBS volume. Termination is irreversible. Stopping an instance would retain its EBS disk rather than complete this cleanup.

Retained day3-vpc-lab, both subnets, the route tables and associations, Internet Gateway, security group, and key pair for the next session. Basic VPC networking has no ongoing resource charge. Charges already incurred during testing still apply; this report is not an account-wide billing audit.

8. Test results and remaining work

Test / task	Result
VPC, subnet and public routing setup	Completed
Security group and key pair	Completed
Public EC2 SSH access	Passed
Outbound HTTPS to AWS website	Passed: HTTP/2 200
Public-to-private instance connectivity	Not tested
Private-subnet internet access via NAT	Not tested; NAT not deployed
EC2 termination and root disk cleanup	Completed

Next session

Reuse the retained network. Review access methods and required security rules before adding a private-subnet test instance. Record the private connectivity results separately. Deploy NAT only if intentionally needed and after reviewing its costs.

Publishing checklist

Before committing documentation to GitHub, redact account IDs, personal public IPs, and identifiers you do not want public. Never include PEM contents, AWS access keys, passwords, or session tokens. Keep the distinction between tested configurations and conceptual discussion.